

Spiking Neuron Models

-Izhikevich single neuron model-

-Fitzhugh-Nagumo model-

Brain Modeling Project

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1. Introduction

Spiking neuron models are widely used to study neural dynamics because they capture key behaviors such as excitability, bursting, and adaptation while remaining computationally efficient. In this project we extended two models from the Laboratory 2 of the Brain Modeling class, the **Izhikevich single neuron model** and the **FitzHugh-Nagumo model**, with analysis tools and reusable experiment components to make simulation easier to run, compare, and interpret.

For the Izhikevich model, we extended it through reusable experiments and quantitative analysis: returning the recovery variable u together with spike times, reusable stimulus generators (step, pulses, ramp, sine), spike detection utilities (spike times, inter-spike intervals (ISI), firing rate), f-I curve generation for any neuron type, phase-plane plotting ($v - u$), and compact multi-type comparison figures under identical stimulation. For the FitzHugh-Nagumo model, extensions are done through equilibrium and stability analysis by computing nullcline intersections, evaluating the Jacobian and eigenvalues at equilibria to classify stability, using a modern integrator (*solve_ivp*) with support for time-varying input currents, and producing f-I curves via sweeps of I_{ext} . Together, these extensions transform the baseline models into a small, consistent framework for reproducible experiments and model comparison.

2. Background

2.1 The Izhikevich model

The **Izhikevich model** is a simple but powerful **spiking neuron model** that can reproduce many known firing patterns of cortical neurons, including **regular spiking** and **bursting**. It combines the **biological realism** of Hodgkin–Huxley–type models with the **computational efficiency** of integrate-and-fire models. Because of its low computational cost, it is possible to simulate large populations of neurons in real time on a standard computer. A key advantage of this model is that it uses only **four parameters**, which makes it easy to tune and analyze.

The model is described by a **two-dimensional system of differential equations**:

$$v' = 0.04v^2 + 5v + 140 - u + I$$

$$u' = a(bv - u)$$

together with an **after-spike reset rule**:

$$\text{if } v \geq 30mV, \text{ then } \begin{cases} v \leftarrow c \\ u \leftarrow u + d \end{cases}.$$

Here, v and u are dimensionless variables, and a , b , c , and d are dimensionless parameters, and $' = d/dt$, where t is the time. The variable v represents the **membrane potential** and u is a slower **recovery variable** that models the effects of ionic currents, such as potassium activation and sodium inactivation. This recovery variable provides negative feedback to the membrane potential and is responsible for **adaptation and bursting behavior**.

The four parameters control the qualitative **firing dynamics** of the neuron. Parameter a sets the time scale of recovery, b determines how strongly the recovery variable reacts to voltage changes, c defines the voltage reset after a spike, and d controls the strength of spike-frequency adaptation. By adjusting these parameters, the same compact model can reproduce a wide range of neuronal behaviors, including regular spiking, fast spiking, and bursting patterns.

2.2 The FitzHugh-Nagumo model

The FitzHugh-Nagumo (FHN) model is a two-dimensional simplification of the Hodgkin-Huxley model. It captures the essential mathematical properties of excitation and propagation of nerve impulses—such as the existence of a threshold, refractory period, and train of spikes—while remaining analytically tractable due to its reduced dimensionality.

The dynamics of the neuron are described by a system of two coupled differential equations:

$$\begin{aligned} \frac{dv}{dt} &= v - \frac{v^3}{3} - w + I_{ext} \\ \frac{dw}{dt} &= \frac{1}{c}(v + a - bw) \end{aligned}$$

In this model, v represents the membrane potential and acts as the fast voltage variable, while w is a slower recovery variable that provides negative feedback, abstracting ionic mechanisms like potassium activation and sodium inactivation. The term I_{ext} denotes the external input current applied to the neuron. The system's behavior is further controlled by the parameter c , which sets the time scale separation ensuring w evolves much slower than v , and by constants a and b , which determine the specific shape and intersection of the nullclines.

Nullclines and Phase Plane Analysis

A major advantage of the FHN model is that it allows for clear phase plane analysis. The system’s behavior is driven by its **nullclines**, which are simply the curves where the rate of change for each variable drops to zero:

v-nullcline ($\dot{v} = 0$): This is a cubic curve described by

$$w = v - \frac{v^3}{3} + I_{ext}$$

w-nullcline ($\dot{w} = 0$): This corresponds to a linear function described by

$$w = \frac{v + a}{b}$$

The points where the curves intersect represent the system’s **fixed points**. The stability of these points is crucial because it dictates the neuron’s behavior: whether it stays at a resting potential or enters a **limit cycle** (oscillatory spiking).

When we vary the external I_{ext} we are essentially shifting the cubic nullcline vertically. This shift changes the intersection point’s location and stability, potentially triggering a switch from silence to repetitive firing through a mechanism known as **Hopf bifurcation**.

3. Methods / Implementation

3.1 Shared utilities

To enable systematic and reproducible experiment, we implemented a **modular simulation pipeline** that separates **stimulus generation**, **neuron dynamics**, and **analysis**. Instead of embedding input currents or measurements directly inside the model code, each component is designed to be **independent** and **reusable**. All experiments follow the same workflow, a stimulus is first defined (as a precomputed current trace $I(t)$ or as a time-dependent input function), the neuron model is then simulated, spike events are detected from the resulting voltage activity, summary statistics (e.g., spike count, ISI, firing rate) are computed, and the results are finally visualized using plotting utilities. This design ensures that identical inputs and analysis tools can be applied to different neuron models without modifying the core solver.

3.1.a Stimulus generators

Within this framework, external inputs are handled through **dedicated stimulus generators** rather than hard-coded currents inside the neuron equations. For the Izhikevich model, stimuli are produced as precomputed numpy arrays aligned with the simulation time grid, including **step currents**, **pulse trains**, **ramp inputs**, and **sinusoidal signals**. Each generator returns an $I(t)$ vector that can be directly passed to the solver, enabling rapid testing of different protocols without modifying the model implementation. For the Fitzhugh-Nagumo model, stimuli are

supported both as constant parameters and as time-dependent functions through the *solve_ivp* interface, allowing inputs such as **steps** or **sinusoidal currents** to be defined as callable functions $I(t)$. This dual design, array-based for discrete simulations and function-based for continuous integration, decouples stimulation from the neuron dynamics and allows identical input protocols to be reused across models, ensuring consistent comparisons and reproducible experiments.

3.1.b Spike detection

Following simulation, spike detection is performed as a standardized analysis step applied to the voltage traces. For the Izhikevich model, spikes are detected intrinsically through the model’s reset mechanism. Whenever the membrane potential reaches the predefined peak threshold ($v \geq 30mV$), the event is recorded and the corresponding time is stored directly during simulation. In contrast, the FitzHugh–Nagumo model does not include an explicit reset rule. Therefore, spikes are identified numerically by detecting upward threshold crossing of the continuous voltage signal, optionally using linear interpolation to improve temporal accuracy. From the extracted spike times, several quantitative metrics are computed, including spike count, inter-spike intervals (ISI), firing rate, adaptation trends, and burst statistics. This unified spike-time representation allows identical statistical analyses, such as ISI histograms, firing-rate estimation, and f-I curve construction, to be applied consistently to both models.

3.1.c f-I pipeline

Building on the shared stimulus and spike-detection tools, we implemented a unified **frequency-current (f-I) analysis pipeline** that measures how the **firing rate** changes as a function of **input current amplitude**. For both models, the procedure consists of sweeping the external stimulus over a predefined range of amplitudes, running a separate simulation for each value, detecting spike times from the resulting voltage trace, and computing the steady-state firing rate within a specified time window after discarding the initial transient. In the Izhikevich implementation, step currents are generated as arrays and passed to the solver, while in the Fitzhugh–Nagumo model the same protocol is applied using either constant I_{ext} values or time-dependent input functions through the *solve_ivp* integrator. The firing rate is then calculated as the number of spikes per unit time and plotted against the corresponding input amplitude to obtain the f-I curve. Because the procedure is model-independent, it enables direct and fair comparison of excitability properties across neuron types and parameter sets, providing a consistent and reproducible characterization of spiking behavior.

3.2 Izhikevich extensions

3.2.a Returning the recovery variable u

The baseline implementation of the Izhikevich model typically returns only the membrane potential trace $v(t)$. To enable a more complete dynamical analysis, we extended the simulator to also return the recovery variable $u(t)$ together with the detected spike times. The recovery variable represents slow negative feedback associated with ionic currents and is directly responsible for spike-frequency adaptation and bursting behavior. By exposing $u(t)$ as an output of the solver, it becomes possible to directly observe how recovery processes evolve during stimulation, to quantify adaptation effects or burst structure, and to perform phase-plane analysis. The simulator therefore consistently outputs the triplet $(V, u, \text{spike times})$, providing richer information while preserving the original numerical integration scheme.

3.2.b Phase-plane analysis

To better understand the interaction between the fast membrane potential dynamics and the slower recovery process, we implemented phase-plane visualization tools that plot the system trajectory in the (v, u) state space. Instead of observing only voltage as a function of time, the *plot_phase_plane* function shows how the two variables evolve together, providing a clearer view of the underlying dynamics. We also included nullcline plots corresponding to $dv/dt = 0$ and $du/dt = 0$, which highlight equilibrium regions and illustrate how their intersections and relative positions influence the neuron's behavior. Together, these visualizations help to see how changes in the model parameters lead to different firing patterns, such as regular spiking, bursting, or resonator dynamics, and offer a simple graphical complement to standard time-series plots.

3.2.c Multi-type comparison plots

To enable clear and consistent comparison across different firing regimes, we defined a set of predefined parameter configurations representing characteristic neuronal behaviors (regular spiking, intrinsically bursting, chattering, fast spiking, low-threshold spiking, and resonator). A dedicated comparison function runs the simulator for each parameter set using the same stimulus, time discretization, and numerical integration settings, and displays all resulting voltage traces within a single figure. By controlling all experimental conditions except the model parameters, observed differences in activity arise exclusively from changes in the dynamical properties of the neuron. This unified visualization provides an efficient way to compare firing patterns side by side, supporting both qualitative inspection and quantitative evaluation of how parameter variations influence neuronal behavior.

3.3 FitzHugh-Nagumo extensions

3.3.a Equilibrium and stability analysis

To characterize the dynamical landscape of the neuron, we extended the analysis framework to analytically determine equilibrium points and their stability. Instead of only relying on numerical simulation, the code computes the intersection nullclines to locate fixed points and evaluates the Jacobian matrix at these coordinates. By calculating the eigenvalues of the Jacobian for any given input current, we can rigorously classify the equilibrium as stable or unstable. This analytical step provides a direct method to predict the onset of oscillations (Hopf bifurcation) and to map the distinct dynamical regimes-quiescence, spiking, and excitation block before running the time domain simulation.

3.3.b Numerical integration (*solve_ivp*)

To simulate the continuous time evolution of the system with high numerical precision, we utilized the adaptive Runge-Kutta method (RK45) provided by *scipy.integrate.solve_ivp*. This solver automatically adjusts the time step to accurately capture rapid changes in the membrane potential during an action potential while maintaining efficiency during slow recovery phases. Furthermore, we structured the implementation to accept time-dependent input functions rather than just static values. This flexibility allows the model to be tested with dynamic stimuli, such as ramps or sinusoidal currents, ensuring robust analysis of the neuron's response to time-varying inputs.

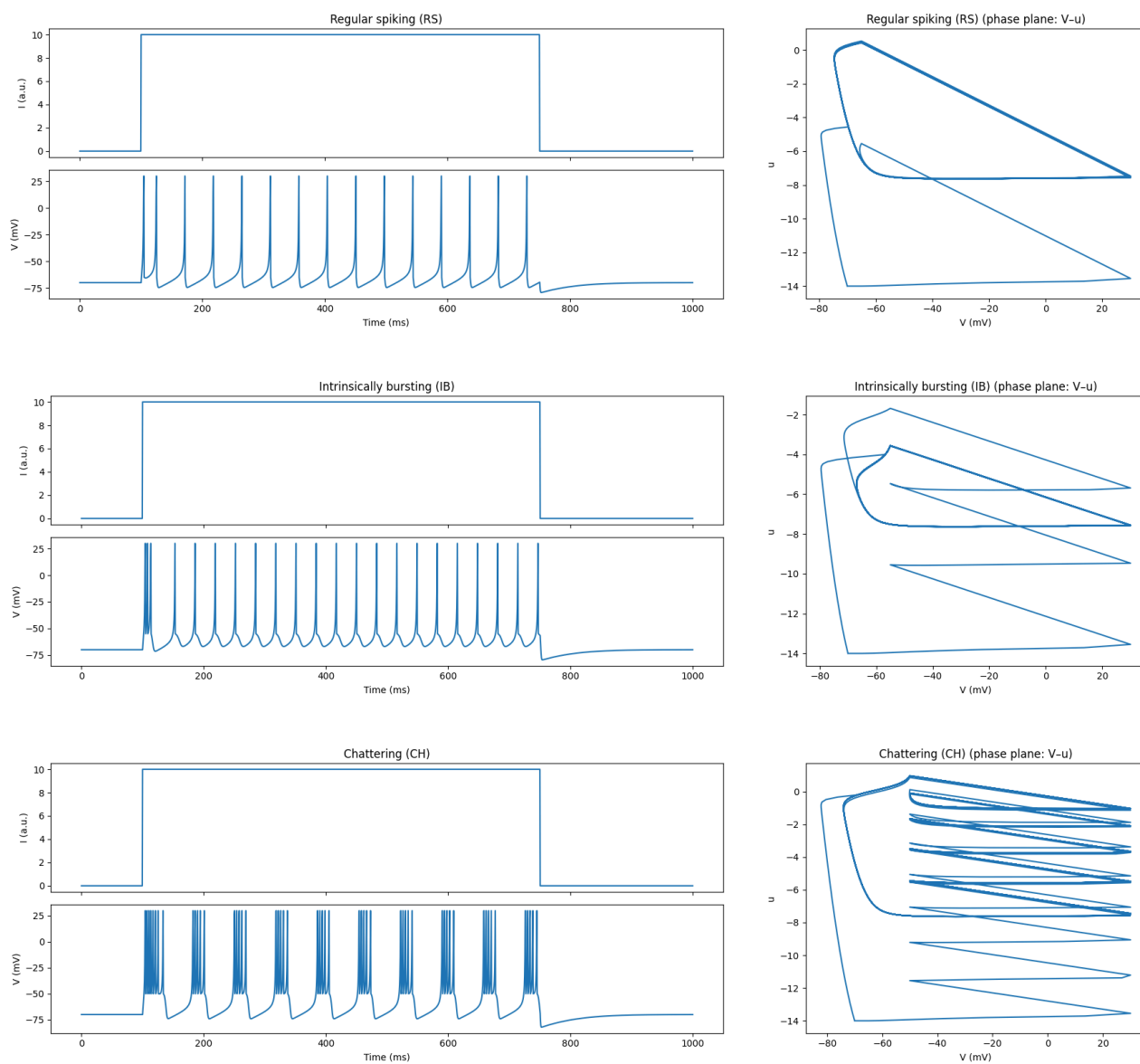
4. Results

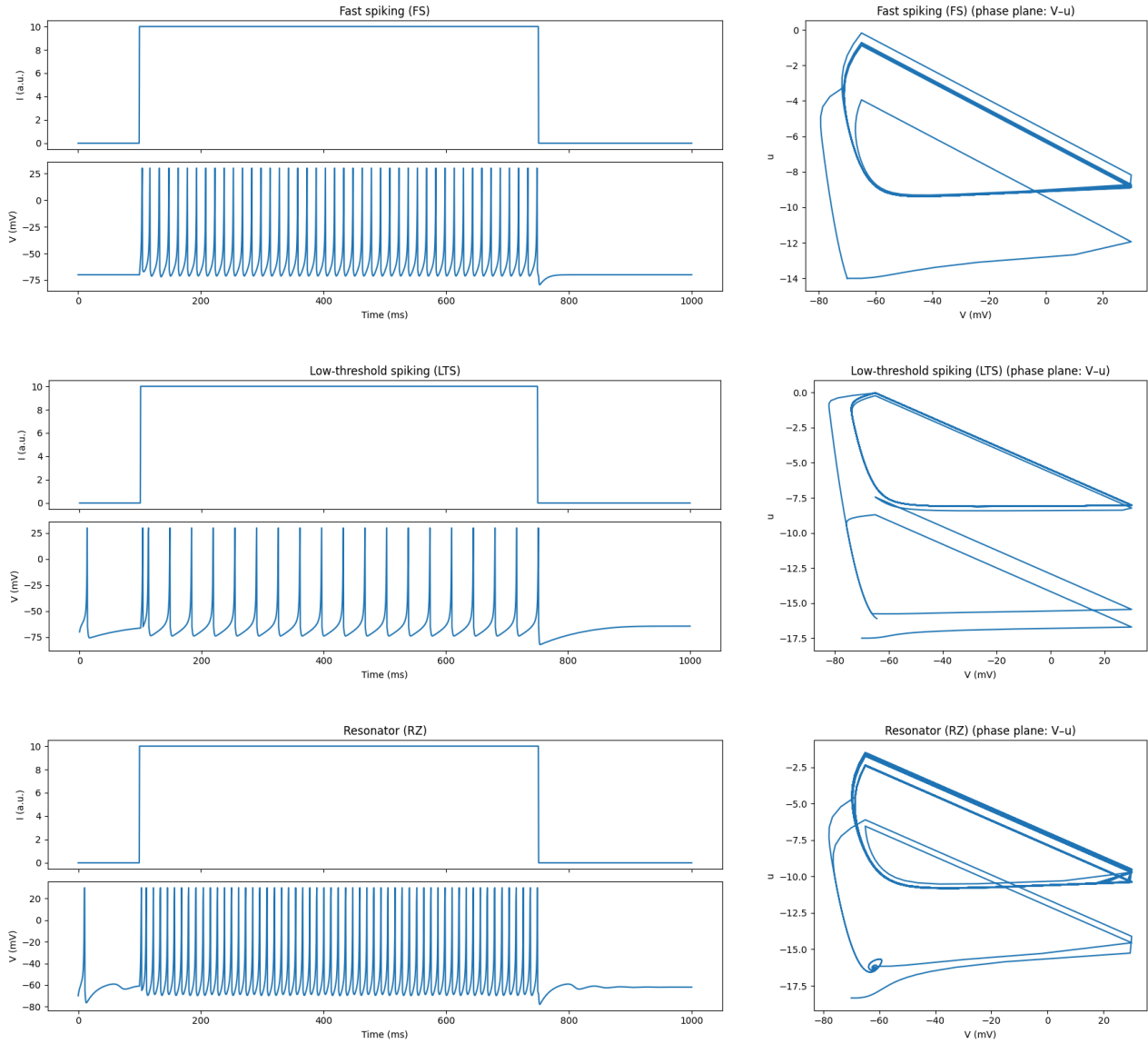
4.1 Izhikevich Model Results

The extended Izhikevich model was tested through a series of structured simulations to study different aspects of neuronal behavior, including excitability, regular, spiking, bursting, underlying dynamics, and the effect of parameter changes. Using the same stimulus protocols and shared analysis tools across all experiments, we measured spike timing, firing rates, phase-plane behavior, and input-output responses. The results show that the model can reproduce several biologically realistic firing patterns while allowing clear and reproducible quantitative analysis.

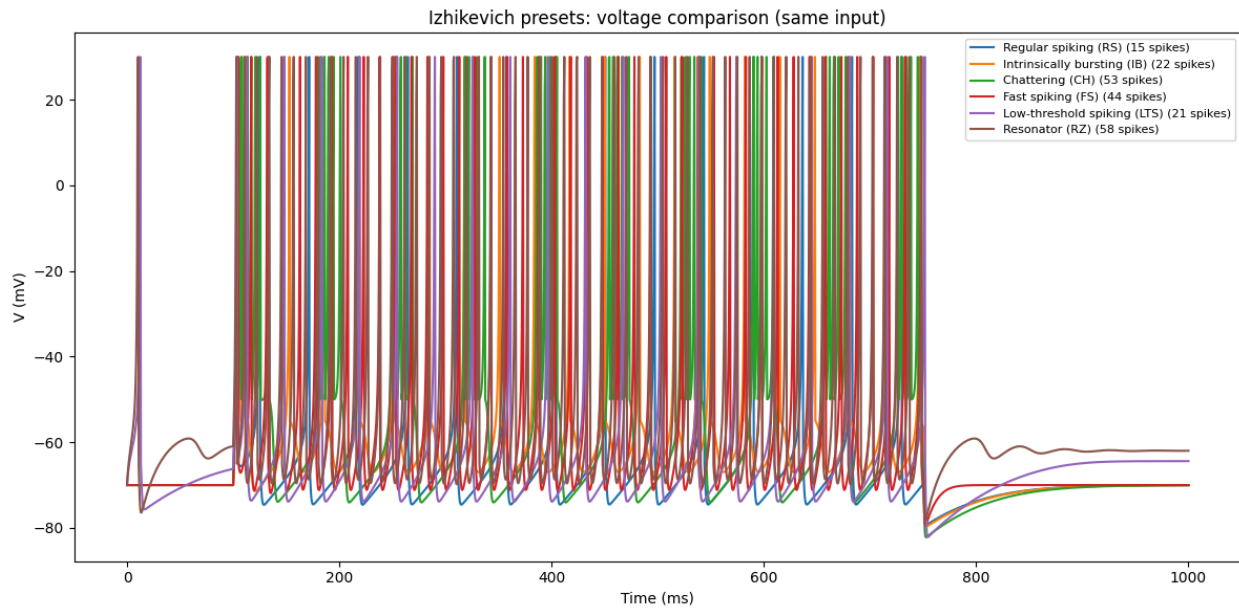
Applying a constant step current revealed distinct activity patterns for each parameter preset. Regular spiking (RS) neurons produced periodic spike trains with nearly constant inter-spike intervals, while intrinsically bursting (IB) neurons generated short initial bursts followed by isolated spikes. Chattering (CH) neurons displayed repeated high-frequency burst packets, fast spiking (FS) neurons fired rapidly with minimal adaptation. Low threshold spiking (LTS) neurons showed clear spike-frequency adaptation, with the firing rate gradually slowing down

over time during continuous stimulation. Finally, resonator (RZ) neurons showed oscillatory dynamics, responding strongly to periodic inputs.



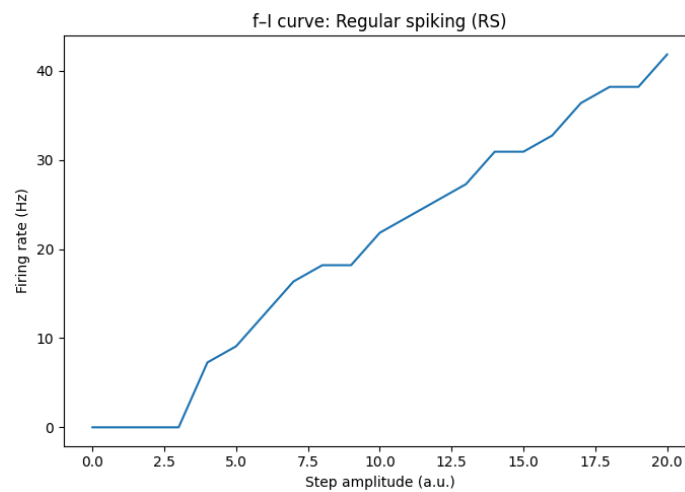


Because all presets were simulated with the same input current, time grid, and solver, differences in activity arise solely from intrinsic parameters. This confirms that the four parameters (a , b , c , d) are sufficient to generate a broad spectrum of firing phenotypes.



4.1.a Extension 1

The excitability of the regular spiking neuron was quantified using the frequency-current (f-I) curve. The input step amplitude was systematically increased while measuring the steady-state firing rate.

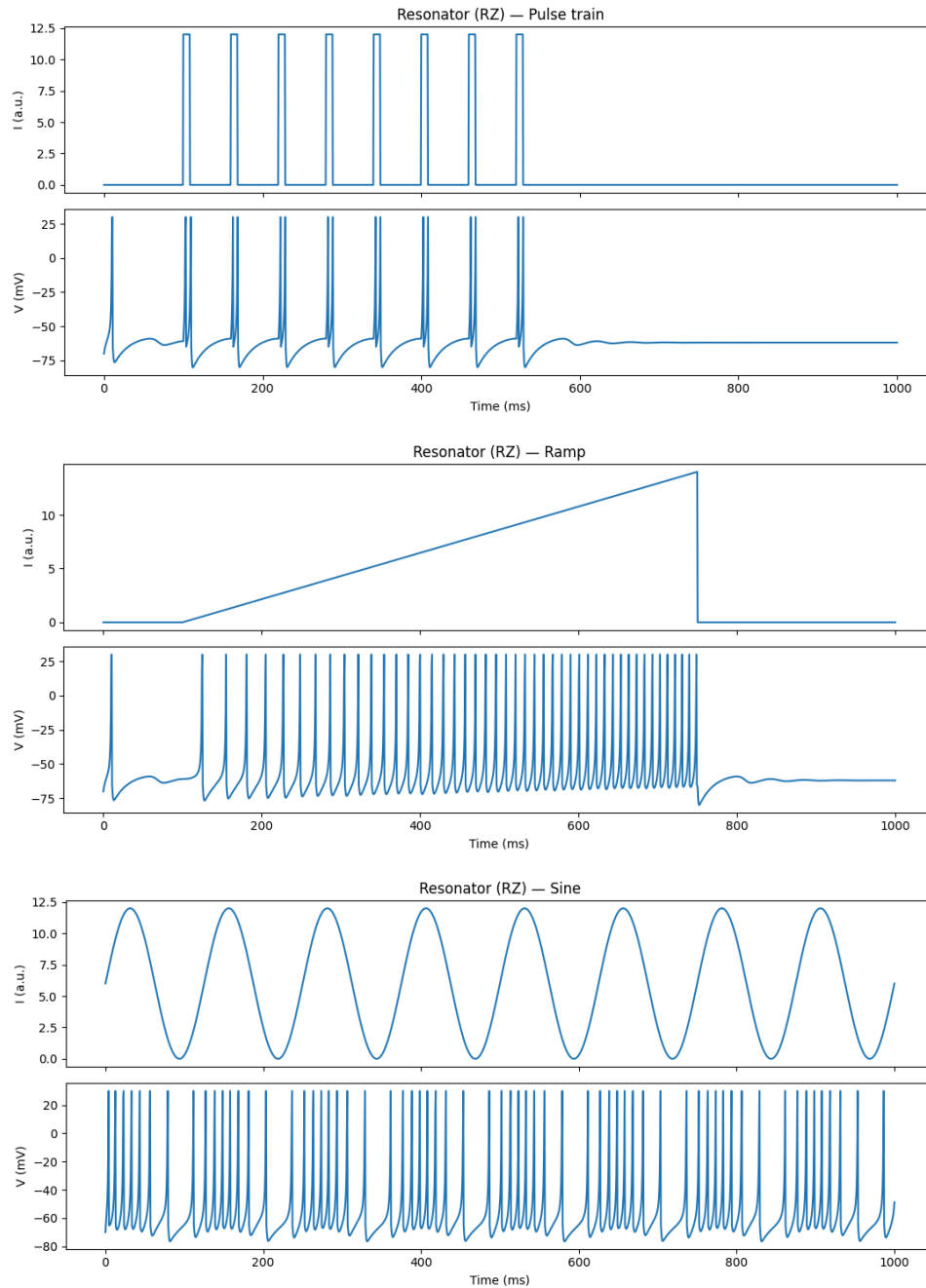


The curve shows a clear firing threshold. The neuron remains silent for small currents and begins spiking only when the amplitude reaches approximately 5-6 a.u.. Beyond this threshold, the firing rate increases approximately linearly with increasing input strength. Once activated, the neuron fires regularly with a stable frequency for each constant current.

The monotonic trend shows how the neuron's firing rate steadily increases as the input current becomes stronger. It confirms that the regular spiking (RS) neuron represents stimulus intensity by changing its firing rate.

4.1.b Extension 2

To evaluate temporal selectivity, the resonator configuration was simulated with pulse trains, ramp current, and sinusoidal inputs.

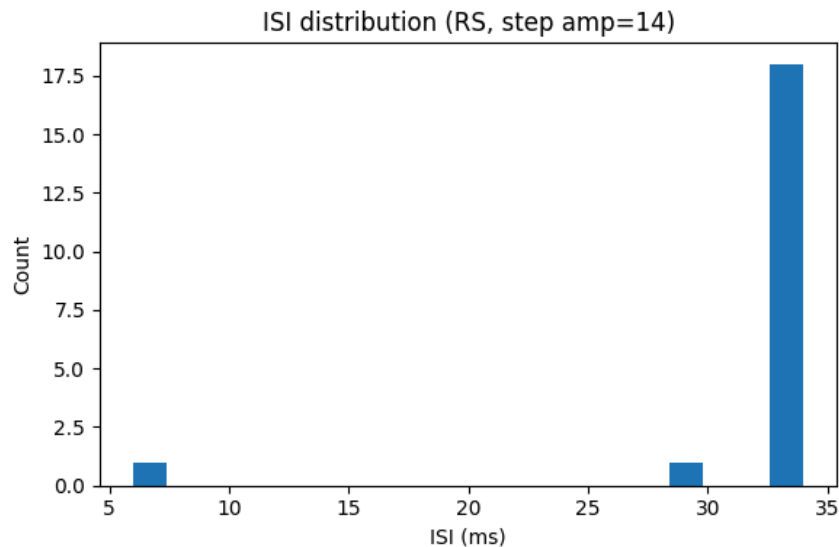


The pulse train induced 17 spikes, indicating that the neuron responded only to a subset of pulses, suggesting temporal filtering or selective responsiveness. The ramp stimulus produced 47 spikes, with firing onset delayed until the input crossed threshold and frequency increasing with amplitude. The sinusoidal input generated 64 spikes and phase-locked activity, demonstrating strong responsiveness to oscillatory stimulation.

This behavior shows that resonator neurons are especially sensitive to rhythmic or periodic inputs, which matches their natural tendency to produce small oscillations even below the spiking threshold.

4.1.c Extension 3

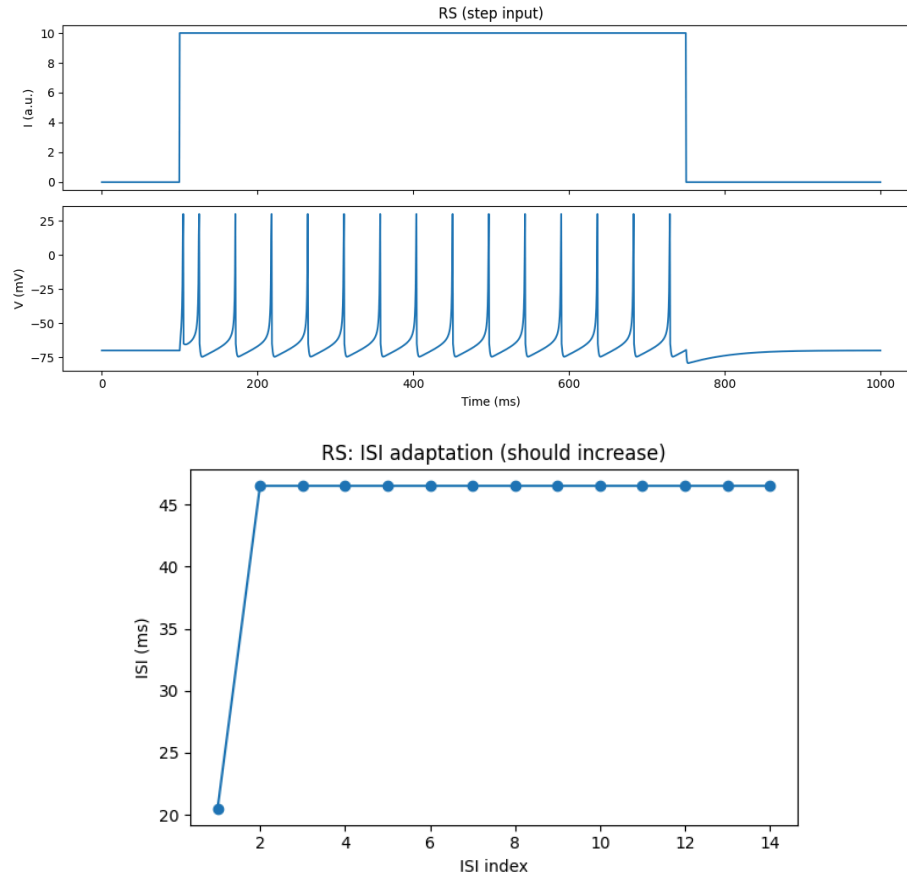
Inter-spike intervals (ISI) were computed to measure firing regularity. For the regular spiking neuron with a strong constant input, 21 spikes were recorded with a mean of ISI of 32.33ms . The ISI histogram shows a narrow peak around this value, indicating highly consistent spike timing and low variability.



This narrow distribution confirms that regular spiking neurons fire in a steady and almost periodic way when the input is constant. The small spread in the histogram is mainly due to short transient effects at the beginning of the simulation.

4.1.d Extension 4

Spike-frequency adaptation was assessed by plotting ISI values against spike index. The adaptation curve shows shorter intervals at the beginning of simulation followed by progressively longer intervals that stabilize over time.



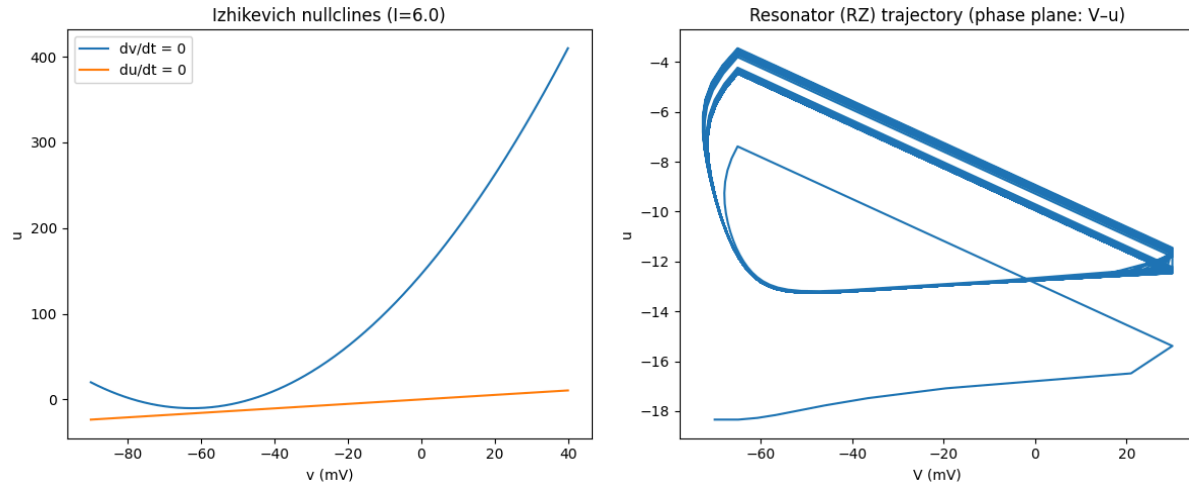
This upward trend indicates that firing slows during sustained stimulation. The effect arises from accumulation of the recovery variable u , which introduces slow negative feedback and reduces excitability after each spike. The results quantitatively demonstrate the role of the recovery process in controlling spike timing.

Burst structure was analyzed automatically using an ISI-threshold-based detector. The intrinsically bursting neuron produced one initial burst of three spikes (22 total spikes), followed by isolated activity. In contrast, the chattering neuron produced sustained bursting throughout the stimulus, generating 10 bursts with an average size of approximately 5 spikes (53 spikes).

Neuron type	Total spikes	Bursts	Mean burst size	Burst pattern
IB	22	1	3.0	single short burst
CH	53	10	5.3	repeated multi-spike bursts

These results clearly distinguish sparse bursting (IB) from high-frequency burst packets (CH) and confirm that the burst detection procedure correctly identifies different bursting regimes without manual inspection.

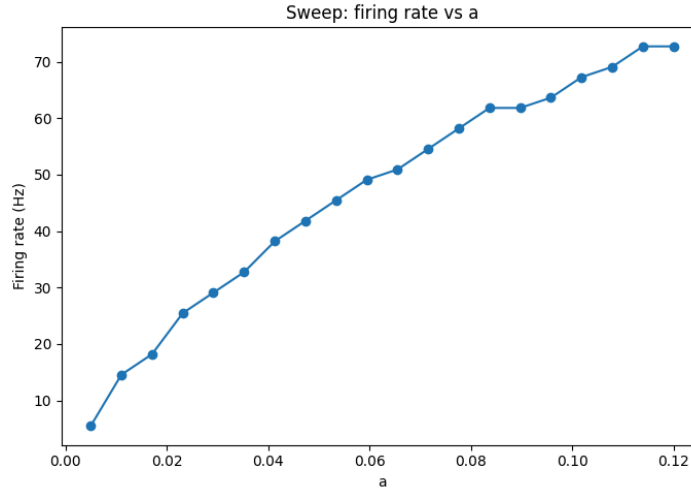
The system dynamics were further examined in the (v, u) phase plane by plotting nullclines and trajectories. The $dv/dt = 0$ curve forms a parabola and the $du/dt = 0$ curve a straight line, whose intersections determine equilibrium points.



For the resonator neuron with constant input, trajectories form stable oscillatory loops (limit cycles), corresponding to repetitive spiking. The position and shape of the nullclines explain how the system moves away from equilibrium and settles into sustained oscillations. This interpretation makes it easier to understand why resonator neurons exhibit oscillatory or frequency-selective behavior.

Phase-plane analysis therefore compliments the voltage time-series plots by providing a clearer picture of the underlying dynamics of the system.

Finally, the influence of intrinsic parameters was examined by sweeping the recovery time-scale parameter a . The resulting firing rate vs. a curve shows an inverse relationship, where as a increases, the firing rate decreases.



Larger values of a cause the recovery variable u to activate more rapidly, strengthening spike-frequency adaptation and lengthening inter-spike intervals. Consequently, the neuron fires less frequently. This confirms the theoretical role of a in controlling adaptation strength and demonstrates that small parameter changes can significantly alter excitability.

Across all experiments, the extended Izhikevich model successfully reproduces diverse neuronal behaviors and supports both qualitative visualization and quantitative analysis. The model shows clear firing thresholds, graded f-I responses, stimulus-dependent dynamics, regular or bursting spike statistics, interpretable phase-plane structure, and predictable parameter sensitivity. These results validate the model as a flexible and computationally efficient tool for systematic characterization of single-neuron dynamics.

4.2 Fitzhugh-Nagumo Model Results

While the Izhikevich model allows for a broad phenomenological survey of firing patterns, the FitzHugh-Nagumo (FHN) model enables a rigorous geometric analysis of the excitability mechanism itself. We focused our investigation on the topological changes in the phase plane that drive the transition from rest to spiking.

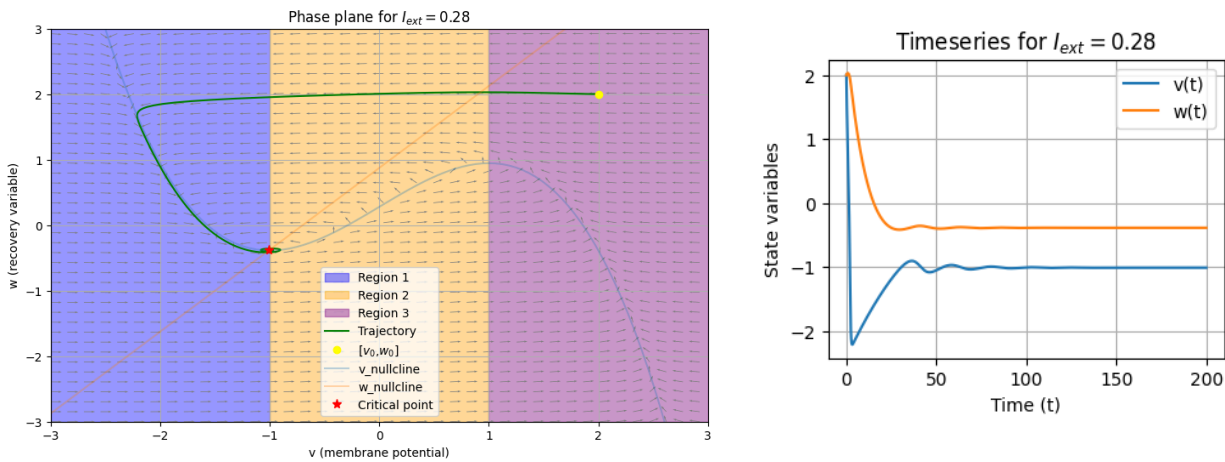
4.2.a Extension 1

To understand the neuron's resting behavior and stability, we analyzed the system under weak stimulation ($I_{ext} \approx 0, 28$). As illustrated in the phase portrait, the cubic v-nullcline (blue) and the linear w-nullcline (orange) intersect on the left stable branch (Region 1). This geometric configuration establishes a single fixed point. By evaluating the Jacobian matrix at this intersection, we determined that the eigenvalues are complex with a negative real part, analytically classifying the equilibrium as a stable focus.

The time-domain simulation confirms this prediction. The trajectory (green line) initiates away from the fixed point and exhibits damped oscillations that spiral inward toward equilibrium. Biologically, this represents subthreshold resonance: the membrane potential fluctuates in response to the input but is ultimately pulled back to the resting potential by the restorative drive of the recovery variable w , failing to cross the threshold required for a full action potential. The time series plot clearly shows these oscillations decaying over time until the system reaches steady state.

Importantly, the phase portrait reveals a complex basin structure around the stable fixed point. The middle branch of the v -nullcline acts as a threshold separatrix: trajectories starting above this branch (like our initial condition $[v_0, w_0] = (2, 2)$) perform a large excursion resembling an action potential before spiraling back to rest. In contrast, trajectories starting closer to equilibrium converge directly without generating a spike-like event.

This threshold behavior is characteristic of Type II excitability: the system does not sustain oscillations at this input level, but initial conditions determine whether the neuron produces a transient spike before returning to rest. At higher currents (*e.g.* $I_{ext} \approx 0.35$), a supercritical Hopf bifurcation would create a true stable limit cycle, enabling sustained repetitive firing.

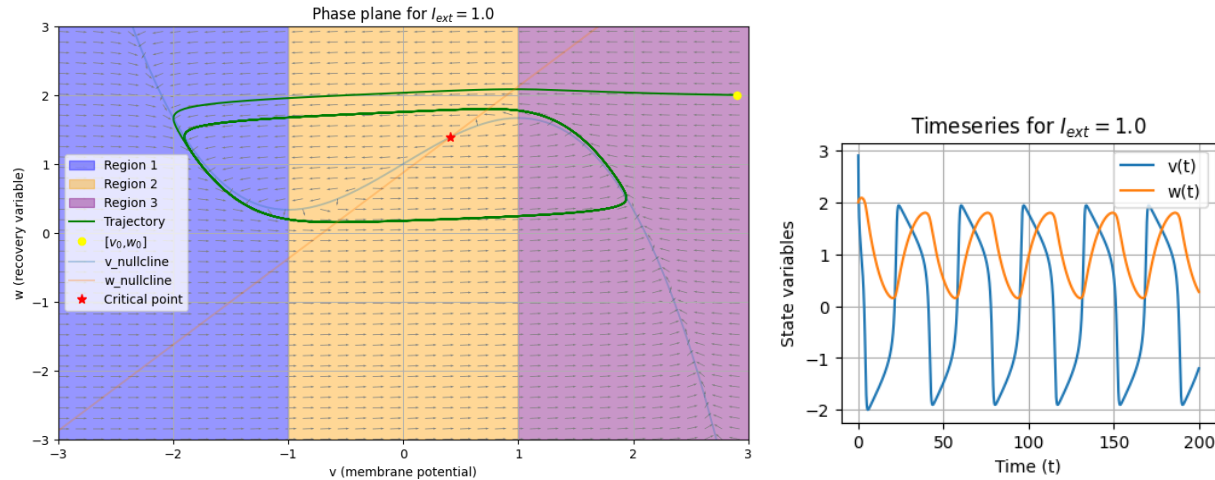


4.2.b Extension 2

We increased the external current to $I_{ext} \approx 1.0$ to drive the system into its active firing regime. The increased current shifts the cubic nullcline vertically, moving the fixed point onto the unstable middle branch. Here, the real part of the Jacobian eigenvalues becomes positive, signifying a loss of stability via a Supercritical Hopf Bifurcation.

With the fixed point now acting as a repeller, the system is forced into a global attractor: a stable limit cycle. The phase plane reveals the classic relaxation oscillator shape, characterized by long

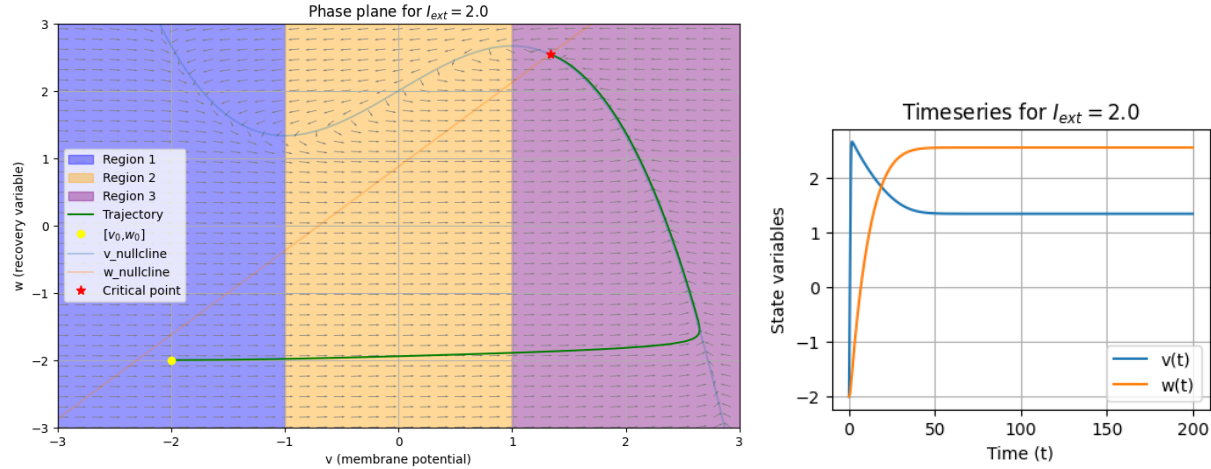
trajectories along the slow nullclines and rapid jumps between them. This corresponds to tonic spiking in the time domain simulation, where the neuron fires continuously at a regular frequency. The voltage trace demonstrates sustained, periodic action potentials driven by the inherent instability of the equilibrium state rather than by noise or stochastic inputs.



4.2.c Extension 3

Finally, we pushed the model to its limits to characterize the cessation of firing and the frequency response. Under high stimulation ($I_{ext} \approx 2.0$), the fixed point shifts again, landing on the right stable branch. The Jacobian analysis confirms a return to stability, classifying the equilibrium as a stable node. The phase plane trajectory performs a single excursion before converging asymptotically to this fixed point.

This behavior demonstrates the depolarization block phenomenon. As shown in the time series simulation, the voltage trace exhibits a single onset spike followed by a permanent high-voltage plateau. The external current is so strong that it saturates the recovery variable, preventing the membrane from repolarizing and locking the neuron in a depolarized state where no further spikes can be generated.

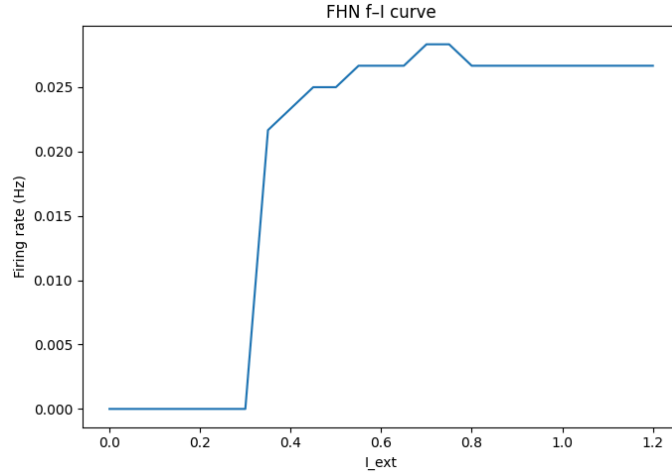


4.2.d Extension 4

To complete the dynamical analysis, we generated the frequency-current (f-I) relationship to determine the neuron's excitability class. In distinct contrast to the continuous, linear frequency increase observed in the Izhikevich model results, our simulations reveal a discontinuous onset characteristic of Type II excitability.

The firing rate remains strictly at zero below a critical threshold ($I_c \approx 0.325$ determined numerically by identifying the Hopf bifurcation point where the real part of the Jacobian eigenvalues changes sign). Immediately upon crossing this value, the system begins oscillating at a distinct non-zero frequency (≈ 0.022 Hz in model time units, corresponding to one spike approximately every 45 time units). This discrete jump serves as the dynamical signature of the supercritical Hopf bifurcation, confirming that the FitzHugh-Nagumo model acts as a resonator.

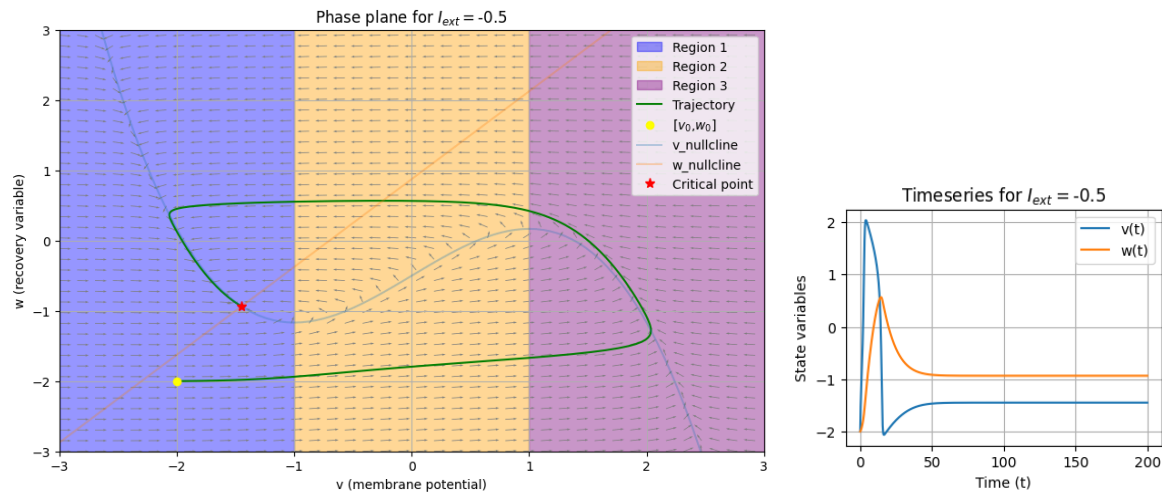
Beyond the bifurcation threshold, the firing frequency increases gradually with input current until it reaches a maximum, after which further increases in I_{ext} drive the system into the depolarization block regime (*Region3*, $I_{ext} > 1.42$), where the neuron stops firing and remains locked in a high-voltage plateau. This non-monotonic f-I relationship is a hallmark of Type II neurons and contrasts sharply with the monotonically increasing f-I curve characteristic of Type I excitability.



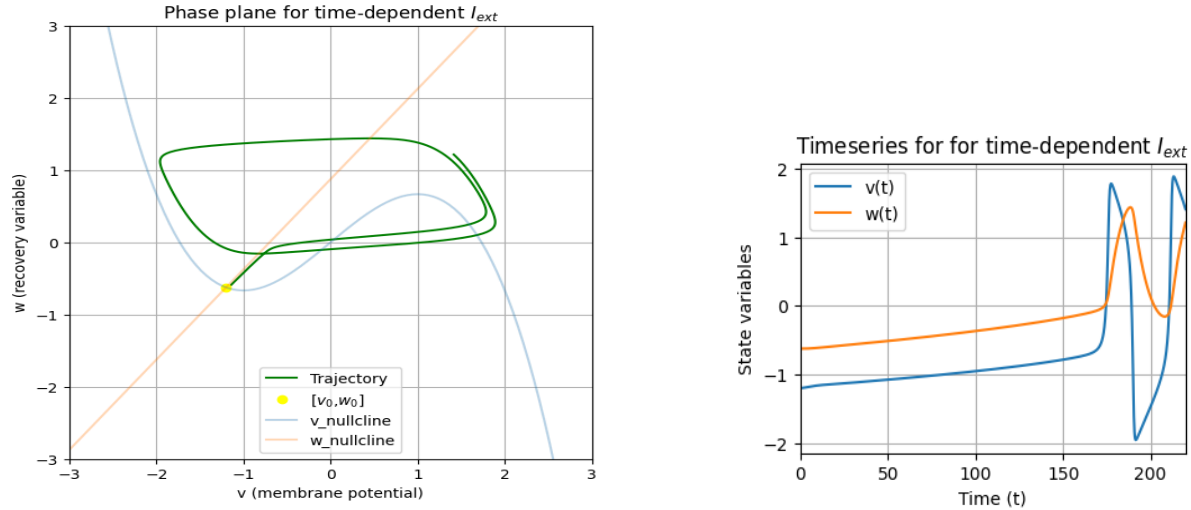
4.2.e Extension 5

To conclude the analysis of the FitzHugh-Nagumo model, we investigated its response to time-varying and inhibitory stimuli to verify complex biological properties beyond simple tonic spiking. First, we characterized the system's behavior under a constant inhibitory current ($I_{ext} = -0.5$).

As illustrated in the phase portrait, this negative drive shifts the fixed point, marked by a red star, deep into the left stable branch of the cubic nullcline. The corresponding time-series simulation confirms that the system, starting from its initial conditions, generates a single onset spike before decaying toward a hyperpolarized steady state. Biologically, this transient excursion demonstrates the non-linear nature of the membrane's return to equilibrium; while the inhibitory drive eventually locks the neuron in a resting state, the initial dynamics allow for a singular activation before the restorative variable w dominates the trajectory. This stabilization at a low-voltage coordinate confirms the robustness of the resting state under sustained hyperpolarization.



Furthermore, we examined the property of neural accommodation by subjecting the model to ramped external stimuli. This reveals the complex non-autonomous nature of the system, where a time-dependent current $I_{ext}(t)$ causes the v -nullcline to shift its vertical position continuously, effectively pushing the state of the system through different regions of the phase plane. In the case of the ramp stimulus ($I_{ext} = 0.003t$), the trajectory initiates near the resting equilibrium and, driven by the increasing input, performs two large excursions reaching $v \approx 2$ and $w \approx 1.4$. These loops represent distinct action potentials generated as the system fails to remain at a fixed point despite the restorative drive. The transitions are marked by characteristic rapid horizontal jumps in the membrane potential followed by a slower recovery phase along the w -nullcline. These observations confirm that the FitzHugh-Nagumo model acts as a resonator sensitive to the derivative of the stimulus (dI/dt), a hallmark of Type II excitability that allows the neuron to act as a high-pass filter, ignoring gradual environmental shifts while remaining responsive to rapid transitions.



5. Discussion

In this project, we analyzed two fundamental frameworks for modelling neuronal excitability. Our results highlight a clear trade-off between phenomenological versatility and dynamical transparency.

The **Izhikevich model** proved superior in reproducing a diverse range of biological firing patterns. As shown in the results, by simply adjusting the four dimensionless parameters (a, b, c, d), the model successfully simulated Regular Spiking (RS), Intrinsically Bursting (IB), and Chattering (CH) behaviors. Its hybrid nature combining continuous differential equations with a discrete reset allows it to capture complex dynamics like spike-frequency adaptation and bursting without the computational cost of calculating ion channel conductances. The f-I curve

analysis for the RS parameter set revealed **Type I excitability**, where the firing rate increases continuously from zero, allowing the neuron to encode stimulus intensity with high precision at low frequencies.

In contrast, the **FitzHugh-Nagumo (FHN) model** provided a deeper geometric insight into the mechanism of excitability. While it lacks the flexibility to model bursting or chattering, its continuous 2D structure allowed us to perform a rigorous phase plane analysis. We successfully visualized how the intersection of nullclines determines the system's state, identifying the transition from quiescence to tonic spiking via a **Hopf bifurcation**. Crucially, our f-I analysis characterized the FHN model as exhibiting **Type II excitability**, marked by a discontinuous onset of firing at a specific non-zero frequency. Additionally, the FHN model captured the phenomenon of **excitation block** at high currents (Region 3), a feature that simpler integrate-and-fire models often fail to represent natively without added constraints.

6. Limitations

Both models have inherent limitations relative to biophysical realism:

- **FitzHugh-Nagumo:** Its primary limitation is the lack of biological mapping for its variables; v and w are abstract quantities, making it difficult to fit experimental voltage traces quantitatively. Furthermore, the model is restricted to single-spike or tonic modes and cannot inherently produce bursting patterns without significant modification (e.g., adding a third variable).
- **Izhikevich:** While computationally efficient for large-scale simulations, the model's "hard reset" is an artificial discontinuity that simplifies the complex biophysics of repolarization. This mathematical hybridity makes it more challenging to analyze using standard dynamical systems theory compared to the smooth vector fields of the FHN model.

7. Conclusion

In this project, we extended and analyzed the Izhikevich and FitzHugh-Nagumo spiking neuron models using a shared and modular simulation framework with reusable stimuli, spike detection, and analysis tools. This structure allowed us to run consistent experiments and compare neuronal behaviors in a systematic and reproducible way.

The Izhikevich model successfully reproduced a wide range of firing patterns, including regular spiking, bursting, and resonator dynamics, and enabled quantitative analysis through ISI

statistics, burst detection, phase-plane plots, and f-I curves. The FitzHugh-Nagumo model, although less flexible in generating firing types, provided clearer insight into the underlying dynamics through nullclines, equilibria, and stability analysis.

Overall, the two models complement each other. The Izhikevich model is efficient and versatile for simulating realistic behaviors, while the FitzHugh-Nagumo model is useful for understanding the mathematical mechanisms of excitability. Together, they provide a simple but effective framework for studying single-neuron dynamics.